

Syed Shakib Shahriar Islam

Jashore - 7408, Bangladesh

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Profile Summary

Electrical and Electronic Engineering graduate with strong academic standing and research experience in AI-driven embedded systems, and biomedical signal processing. Possesses practical, applied experience in scientific programming (Python, C/C++). IEEE student leader with proven technical communication skills. Seeking opportunities to advance career in academic research, and teaching.

Research Interests

Biomedical signal processing, computational modeling, AI-driven embedded systems, intelligent control systems.

Education

Bachelor of Science in Electrical and Electronic Engineering (EEE) 2026

Jashore University of Science and Technology

CGPA: 3.89 / 4.00

Rank: 1st

Higher Secondary Certificate (HSC) 2020

Notre Dame College, Dhaka

GPA: 5.00 / 5.00

Award: Board Merit Scholarship (General), Dhaka Board

Secondary School Certificate (SSC) 2018

Bir Uttam Shaheed Samad School and College, Rangpur Cantonment, Rangpur

GPA: 5.00 / 5.00

Award: Board Merit Scholarship (Talent Pool), Dinajpur Board

Undergraduate Thesis

Development of a Resource-Efficient Surface Electromyography Driven Prosthetic Arm Prototype Using Edge-Deployed Deep Learning

Supervisor: Dr. Md. Moznuzzaman

Developed and validated a 3D-printed transradial prosthetic arm prototype controlled via three-channel surface electromyography (sEMG) array. Collected multi-session, posture-varied data from 20 subjects from the university community. Implemented a deep neural network for real-time gesture recognition, with embedded deployment on ESP-based hardware for low-latency control aimed at rehabilitation assistance.

Publications

- **S. S. S. Islam**, M. R. Hasan, M. Moznuzzaman. “Development and Able-Bodied Validation of a Resource-Efficient Three-Sensor sEMG Prosthetic Arm Prototype Using Edge-Deployed Deep Learning.” *To be submitted: PLOS ONE*.
- M. S. Basunia, **S. S. S. Islam**, et al. “Privacy Preserving Federated Learning for Early Student Dropout Prediction.” *Presented at: 2026 IEEE 5th RAAICON*.

Academic Presentations

Smart Electric Wheelchair with Regenerative Braking

Nov 2025

Poster Presentation – Symposium on Energy Security & Fuel Optimization, JUST

- Presented a conceptual design adapting EV regenerative braking technology for assistive mobility. Modeled cost-benefit analysis and energy recovery curves using Python, demonstrating a 5–10% direct energy recovery and a significant extension of battery runtime.
- **Focus Areas:** Energy Optimization, Concept Modeling, Assistive Technology, Embedded Systems

Projects

AI-Based Vehicle Speed Detection (LimitEye)

Third Year Project

- Developed a custom YOLO-based computer vision model to detect vehicles and license plates in real-time. Utilized LabelImg for dataset annotation and trained the model using Google Colab to implement speed calculation for automatic violation detection.
- **Tools:** Python, LabelImg, Custom Trained YOLO, SORT Algorithm, Google Colab

Industrial Energy Audit: Asian Marble Industries

Energy Management Course

- Conducted a full on-site energy audit of a 118,603 sq. ft. facility, identifying a Power Factor deficiency and proposing an efficiency roadmap projecting a 10–15% cost reduction through PFI upgrades and LED retrofitting.
- **Tools / Skills:** Luxmeter, Tachometer, Energy Auditing, Technical Reporting

Additional Lab Projects

- **Line Follower Robot:** Designed an autonomous robot using an IR sensor array and Arduino PID control for smooth track navigation.
- **DC Motor Control:** Implemented a closed-loop speed control system for DC motors by tuning PID parameters to minimize error and optimize response time for load variations.
- **Overvoltage Protection:** Constructed an Arduino-programmed hardware circuit using a voltage sensor and a 5V relay module to automatically isolate the load during surge events.

Trainings

Industrial Training on Electrical Engineering & Instrumentation – TICI, Narshingdi Jun 2025 – Jul 2025

Training on AI for Immersive Technology (288-hour) – Level 6, NSDA, BDSET Center, Jashore Mar 2025 – Jun 2026

Solar System Planning, Design, Installation and Operation for Engineering Students – Daffodil International University Apr 2026

Leadership & Extra-curricular Activities

IEEE JUST Student Branch

Apr 2023 – Dec 2025

Social Media Coordinator & Lead, Media and Public Relations Wing; Program Coordinator (Previous)

Managed social media platforms, inter-branch communication, publications, and public relations. Hosted and moderated multiple technical webinars and represented the branch in IEEE Bangladesh Section activities.

Munshi Meherullah Hall, JUST

Nov 2024 – Apr 2025

Dining Manager

Directed daily food supply logistics and vendor coordination for a residential hall serving 900 students. Handled financial budgeting, expenditure tracking, and real-time service quality resolution.

Conference Experience

2nd International Conference on Engineering and Technology for Sustainable Development May 2026

Assisted in session conduction, oral presentation evaluation, speaker coordination, and decoration.

International Conference on Emerging Technologies for Sustainable Development Feb 2025

Assisted in registration, session management, and participant communication.

Technical Skills

Python, MATLAB, C/C++, Machine Learning, Arduino & ESP, Proteus, PowerFactory, Dsch2, Microwind, 3D Printing, MS Office, L^AT_EX, Overleaf

Languages

English (Fluent), Bangla (Native), Hindi (Conversational)

References

Dr. Md. Moznuzzaman

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Engineering, JUST

Thesis Supervisor

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Dr. Md. Abdur Rauf Sarkar

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ogy, JUST

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